

Press release

Berlin, April 19, 2022

Siemens Energy presents innovative pole-mounted dry-type transformer for American market

- Launch of single-phase dry-type transformer for pole applications
- More reliable, durable and environmentally friendly alternative to oil-filled transformers

With CAREPOLE™ Siemens Energy launches an innovative dry-type single-phase transformer for pole applications. Designed for the technological requirements of the American grid, the new cast-resin distribution transformer provides a reliable and sustainable alternative to oil-filled transformers.

In American markets, single phase transformers are often installed in the distribution grid, where – as the last step in the power transmission chain – they step down the voltage to household voltage levels. In light of increasing weather and climate extremes, such as heat waves that cause wildfires, the design of new outdoor-installed assets needs to be increasingly focused on their resilience to withstand environmental hazards. Due to its cast-resin design (without oil), the self-extinguishing CAREPOLE has a high level of fire safety, cannot explode, and reduces environmental risks to a minimum.

“For a successful energy transition, we also must rethink the power grid with a focus on making it greener and strengthening its resilience. With our pole-mounted dry-type transformer, we are not just closing a gap in the American market but also presenting a maintenance-free alternative to oil-filled transformers that allows the more environmentally friendly, more reliable, and cost-efficient operation of the distribution grid,” says Eduardo Terzi, Senior Vice President Non-Switching Products at Siemens Energy.

Considering growing energy demand and the fluctuating feed-in of renewable energy, the new transformer was designed to handle high overloads to facilitate power flow balancing and satisfy real time demand at any time. Sealed with cast-resin under vacuum, the compact windings of the

transformer allow for its approximately 20 percent smaller volume size compared to a conventional liquid-filled pole-mounted transformer.

The high mechanical strength of the CAREPOLE's frame and cast resin protects it against corrosion and thereby extends the transformer's service life. At the same time, it prevents copper and aluminum thieves from destroying the transformers and putting themselves and others at risk. The new transformer is designed as a plug-and-play asset with an identical fixing mechanism as fluid-immersed transformers, and is therefore perfectly suited to replace existing installations.

On customer request Siemens Energy's digital transformer solution Sensformer can be added to CAREPOLE to optimize operations and further increase reliability. Equipped with a smart IoT gateway, the transformer can then transmit end-to-end encrypted data to an analytics and visualization platform. Operators benefit from real-time insights on asset health and fleet status. Thanks to these digital features, failures can be localized and rectified as quickly as possible, or often even avoided by identifying operational irregularities at an early stage.

CAREPOLE has a power rating between 10 and 100 kilovolt-amps (kVA), a voltage capacity between 15 and 36 kV and a typical lifetime of over 25 years.

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This press release and a press picture is available at

www.siemens-energy.com/press

For further information on Siemens Energy transformers, please see

<https://www.siemens-energy.com/global/en/offerings/power-transmission/portfolio/transformers.html>

Siemens Energy will present CAREPOLE at IEEE PES T&D Conference & Exposition in New Orleans, April 25 – 28, 2022, at Booth #8728. For further information, please see

<https://www.siemens-energy.com/global/en/news/fairs-events/ieee.html>

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Siemens Energy is one of the world's leading energy technology companies. The company works with its customers and partners on energy systems for the future, thus supporting the transition to a more sustainable world. With its portfolio of products, solutions and services, Siemens Energy covers almost the entire energy value chain – from power generation and transmission to storage. The portfolio includes conventional and renewable energy technology, such as gas and steam turbines, hybrid power plants operated with hydrogen, and power generators and transformers. More than 50 percent of the portfolio has already been decarbonized. A majority stake in the listed company Siemens Gamesa Renewable Energy (SGRE) makes Siemens Energy a global market leader for renewable energies. An estimated one-sixth of the electricity generated worldwide is based on technologies from Siemens Energy. Siemens Energy employs around 91,000 people worldwide in more than 90 countries and generated revenue of €28.5 billion in fiscal year 2021. www.siemens-energy.com.